

5: Polynomial regression:

$$R^2_{\text{adj}} = 1 - \left[\frac{(2 - 0.94) * (10 - 1)}{10 - 3 - 1} \right] =$$

$$1 - \left[\frac{0.06 * 9}{6} \right] =$$

$$1 - \left[\frac{0.54}{6} \right] =$$

$$1 - 0.09 = \underline{0.91}$$

Q7: What is adjusted R^2 ?

Adj R^2 is used to check whether a new feature 'earns its keep' and to compare models with different feature numbers.

We need adj R^2 instead of R^2 for a model like this because if you add predictors to R^2 ~~the~~ it can't fall, so it won't work on models of different sizes